The Point-Class Rank Functional under Birational Wall Crossing: Exact One-Wall Identities toward the $X \times \mathbf{P}^2$ Problem

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Abstract

The Gamma integral structure turns the class of a point into the flat Euler covector which reads ordinary rank. We isolate the behavior of this single row under birational quantum wall crossing. For a smooth projective simple VGIT wall, the formal quantum-D-module comparison of Gu–Yu–Yu sends the common-open point class to the point class in its ambient summand exactly; the assertion concerns the full wall parameter, not only its leading term. For a projective ordinary flop, the Lee–Lin–Qu–Wang continuation preserves the intrinsic point row on each fixed continuation domain. The proof uses support at transverse degree zero and a divisor recursion, and does not invoke full descendant invariance.

We then locate the obstruction to composing discrepant wall comparisons. An incomplete-Gamma connection gives a rank-two Stokes shear from a primitive-sixth formal line into a rank-visible ambient line. The shear persists after imposing a flat integral pairing and a length-three nilpotent tag. Localized Fourier–Laplace transform forgets precisely the constant extension responsible for this jump, while a punctual module at a two-coordinate Fourier corner can transform back to a full-support module. Thus formal monodromy, pairing, integrality, nilpotent tagging, and bare boundary support do not determine the point row. We formulate the remaining two-wall problem as a factorwise rank-zero-target condition and prove that this condition is sufficient for factorization-level point-row invariance. A single signed punctual boundary coefficient is only a candidate shadow of that condition: identifying the two requires an additional marked comparison theorem. For a smooth cubic threefold X, this is the unresolved assembly step in the point-row approach to the $m = 2$ stabilization problem for $X \times \mathbf{P}^2$; no irrationality statement for that product is asserted here.

Keywords. Gamma class; quantum cohomology; birational wall crossing; VGIT; ordinary flop; Stokes matrix; Fourier–Laplace transform; GKZ system.

1 Introduction

Let Y be a smooth projective complex variety. Iritani's Gamma integral structure associates a flat section $s_Y(E)$ of the quantum connection to a topological K-class E and identifies the flat pairing with the Euler pairing [Iri23]. If $p \in Y$ is a point, then

$$[s_Y(E), s_Y(\mathcal{O}_p)) = \chi(E, \mathcal{O}_p) = \mathrm{rk}(E). \tag{1.1}$$

The second entry in (1.1) is therefore a distinguished flat covector. We call it the *Gamma point row*.

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Birational quantum comparison theorems often split a quantum D-module into an ambient summand and summands associated with the wall or blow-up centre. For rank questions, it is unnecessary to identify every entry of the comparison matrix. It is enough to know that the point row restricts to the point row on the ambient summand and vanishes on the supported summands. This paper separates the exact statements of that form from the additional analytic input needed to compose several discrepant walls.

The motivating stabilization problem is concrete. If X is a smooth cubic threefold, the case $m = 2$ asks whether $X \times \mathbf{P}^2$ can be rational. The point-row approach would compare the primitive-sixth formal-monodromy packet of this product with that of projective space along a smooth birational factorization. The one-wall identities proved below supply the local rank row needed by that comparison. What is not supplied is a wall-independent sectorial realization at an intermediate chamber. We call this the $X \times \mathbf{P}^2$ assembly obstruction. The terminology names the missing comparison, not a proved irrationality obstruction for the product.

Our first result concerns a smooth projective simple VGIT wall $X_- \dashrightarrow X_+$ in the sense of Gu–Yu–Yu [GYY25]. Their comparison is a formal, pairing-preserving decomposition

$$\mathrm{QDM}(X_-) \cong \mathrm{QDM}(X_+) \oplus \bigoplus_j \mathrm{QDM}(S)_j \tag{1.2}$$

over an exceptional Laurent completion, where S is the wall quotient. Choose corresponding points $p_\pm$ in the common open set.

Theorem 1.1 (Common-open point column)**.** *At the extremal specialization used in the Gu–Yu–Yu comparison, the ambient coordinate of the point class is exact:*

$$\mathrm{pr}_{X_+} \Phi([p_-]) = [p_+]. \tag{1.3}$$

No assertion is made that every wall coordinate of $\Phi([p_-])$ vanishes.

The proof uses the distinguished master-space lift of the common point. Its wall restriction vanishes. Applying the negative-chamber Fourier isomorphism to the lift and to its wall-frequency derivative shows that the derivative is zero in the specialized completed source. Fourier covariance then makes the positive point image horizontal. A regular-singular recursion kills its apparent positive wall-parameter tail.

Our second result is analytic but crepant. Let $Y \dashrightarrow Y'$ be a projective ordinary flop covered by Lee–Lin–Qu–Wang [LLQW16]. Fix one of their analytic-continuation domains in the extremal variable.

Theorem 1.2 (Ordinary-flop point row)**.** *The analytically continued point section of Y equals the intrinsic point section of Y' on the fixed continuation domain. Consequently the point row has the same zero or nonzero restriction to every sectorially realized formal-monodromy packet whose chosen realization is transported by the graph gauge on that branch.*

Pure extremal curves lie in the exceptional locus and cannot meet a point in the common open set. This computes the point column exactly at transverse Novikov degree zero. A divisor pulled back from the common contraction then kills the transverse tail coefficient by coefficient. This is a theorem about one distinguished column; it does not promote the ancestor comparison of [LLQW16] to full descendant invariance.

The discrepant case contains a further frame comparison. A formal summand, a sectorial lift of that summand, and the intrinsic large-radius Gamma section are different objects. Gu–Yu–Yu prove (1.2) in a Laurent/formal coefficient ring, while Shen–Shoemaker identify Gamma asymptotic classes at an extremal specialization [SS25]. Passing from these results to an intrinsic Gamma point row requires a common pairing-preserving sectorial realization. We state the corresponding one-wall consequence only under that explicit hypothesis.

The negative results show why this distinction is necessary. A rank-two incomplete-Gamma connection has a nonzero Stokes shear from a ζ_6-formal line into a rank-visible ambient line. The example admits a flat integral hyperbolic doubling and remains after tensoring with $\mathbf{Q}[N]/(N^3)$. Moreover, localized partial Fourier–Laplace transform kills the constant extension which records the shear. In two Fourier variables, the delta module at the common origin is invisible after either localization but transforms back to a constant full-support module. Boundary support in Fourier coordinates therefore points in the opposite direction from the desired rank conclusion.

The surviving two-wall statement is one-sided. Complete ray-ordered corrections must have rank-zero *targets*; their sources and scalar coefficients are irrelevant. Under that hypothesis, a finite factorization preserves the point row by elementary linear algebra. A signed punctual coefficient of an oriented boundary complex is a plausible shadow of these factorwise conditions, but it is not proved equivalent to them here. A single alternating coefficient can vanish by cancellation; a marked blockwise comparison is still required.

Sections 2–4 prove the positive results. Sections 5 and 6 give the countermodels. Section 7 states the conditional assembly theorem, and Section 8 records the exact source and inference boundaries.

2 The point row and three distinct frames

We fix the pairing convention before discussing wall crossing. Let $\mathcal{S}(Y)$ denote a space of flat sections of the quantum connection on a domain where the Gamma framing is defined. Iritani's section associated with $E \in K^0_{\mathrm{top}}(Y)$ has the form

$$s_Y(E) = L_Y(\tau, z) z^{-\mu} z^{c_1(Y)} (2\pi)^{-\dim Y/2} \widehat{\Gamma}_Y (2\pi i)^{\deg/2} \operatorname{ch}(E), \tag{2.1}$$

up to the standard choice of normalization. The flat pairing is

$$[s_1, s_2) := (s_1(\tau, e^{-\pi i} z), s_2(\tau, z))_Y, \qquad [s_Y(E), s_Y(F)) = \chi(E, F). \tag{2.2}$$

See [Iri23, Section 1].

Definition 2.1 (Gamma point row)**.** For $p \in Y$, the Gamma point row is

$$\mathfrak{r}_{Y,p}(v) := [v, s_Y(\mathcal{O}_p)). \tag{2.3}$$

On Gamma-framed K-classes it is ordinary rank:

$$\mathfrak{r}_{Y,p}(s_Y(E)) = \chi(E, \mathcal{O}_p) = \operatorname{rk}(E). \tag{2.4}$$

Putting the point in the first entry instead would introduce the usual dimension sign. We use (2.3) throughout.

Three levels of structure occur in the arguments below.

(i) A *formal packet* is a generalized formal-monodromy or exponential block of the $z = 0$ formal connection.

(ii) A *sectorial lift* is a choice of actual flat subspace with that formal asymptotic expansion on an admissible sector.

(iii) The *intrinsic point section* is the Gamma-normalized flat section fixed by the large-radius fundamental solution.

A positive-z gauge can identify formal quantum D-modules without identifying (iii) across two incompatible Novikov completions. Likewise, two sectorial lifts of the same formal packet can differ by a Stokes shear. The zero or nonzero restriction of (2.3) is therefore not determined by the formal packet alone.

Definition 2.2 (Point-row Boolean)**.** For a sectorially realized formal packet $\mathcal{P} \subset \mathcal{S}(Y)$, put

$$b(Y, \mathcal{P}) = \begin{cases} 1, & \mathfrak{r}_{Y,p}|_{\mathcal{P}} \neq 0, \\ 0, & \mathfrak{r}_{Y,p}|_{\mathcal{P}} = 0. \end{cases} \tag{2.5}$$

The paper's exact wall identities either identify the intrinsic point section directly or remain within one specified receiver. Whenever an additional sectorial realization is required, it is stated as a hypothesis.

3 The exact point column at a simple VGIT wall

Let $X_- \dashrightarrow X_+$ be a simple VGIT wall crossing satisfying the hypotheses of [GYY25, Definition 3.2]: the chamber quotients and wall quotient S are smooth and projective, and the normal weights at the wall are ± 1. Assume $r_+ < r_-$ and write $\nu = r_- - r_+$. Theorem 6.2 of [GYY25] gives a pairing-preserving isomorphism

$$\Phi : \mathrm{QDM}(X_-)^{\mathrm{La,red}} \longrightarrow (\tau_+^{\mathrm{red}})^* \, \mathrm{QDM}(X_+)^{\mathrm{La,red}} \oplus \bigoplus_{j=0}^{\nu-1} (\zeta_j^{\mathrm{red}})^* \, \mathrm{QDM}(S)^{\mathrm{La,red}}. \tag{3.1}$$

Its coefficient ring is Laurent in a fractional wall variable and complete in the remaining Novikov and bulk variables. We use only the extremal specialization of (3.1).

Let $p_\pm$ be corresponding points in the common open set. Gu–Yu–Yu's Lemma 3.27 constructs a unique master-space lift a_p with

$$\kappa_+(a_p) = [p_+], \qquad \kappa_-(a_p) = [p_-], \qquad a_p|_{F_0} = 0. \tag{3.2}$$

The last equality follows from the proof of that lemma: the wall restriction is the polynomial determined by the restriction of $[p_+]$ to the positive exceptional locus, which is zero.

Theorem 3.1 (Exact ambient point coordinate)**.** *After setting the master Novikov and bulk variables to zero while retaining the wall variable, the ambient coordinate of (3.1) satisfies*

$$\mathrm{pr}_{X_+} \Phi([p_-]) = [p_+]. \tag{3.3}$$

Proof. Let S denote the wall variable and λ the equivariant wall frequency. Lemma 5.10 of [GYY25] applies to a_p because its wall restriction is zero. It gives

$$\mathrm{FT}_{X_-}(a_p) = [p_-] \tag{3.4}$$

at the specialization under consideration. Lemma 5.8 gives only

$$\mathrm{FT}_{X_+}(a_p) = [p_+] + O(S). \tag{3.5}$$

Apply Lemma 5.10 again to λa_p. Its wall restriction is zero and

$$\kappa_-(\lambda a_p) = \kappa_-(\lambda)[p_-] = 0 \tag{3.6}$$

by degree. The completed source in Proposition 5.2 is finite free, and the negative Fourier map in Theorem 5.5 and Proposition 5.9 is an isomorphism over that completed ring. Base change to the extremal specialization preserves the isomorphism. Equations (3.6) and Lemma 5.10 therefore imply that λa_p is zero in the specialized completed source. In particular,

$$\mathrm{FT}_{X_+}(\lambda a_p) = 0. \tag{3.7}$$

Fourier covariance, Proposition 4.21 of [GYY25], rewrites (3.7) as

$$\left(zS\partial_S + D \star_{\tau_+(S)} + (S\partial_S \tau_+(S))\star_{\tau_+(S)}\right) \mathrm{FT}_{X_+}(a_p) = 0. \tag{3.8}$$

Here D is the wall divisor and $\tau_+(S)$ is the specialized receiver coordinate. Write

$$\mathrm{FT}_{X_+}(a_p) = [p_+] + \sum_{k>0} S^k v_k, \qquad \tau_+(S) = f(S)\mathbf{1} + \eta(S), \tag{3.9}$$

where $f(0) = 0 = \eta(0)$ and $\eta(S)$ has positive cohomological degree. Lemma 5.8 gives $v_k \in H^*(X_+)[z]$. Every nonconstant stable map in a positive multiple of the wall fibre is contained in the exceptional locus and misses p_+. Thus the positive-wall quantum corrections and $\eta(S)$ annihilate $[p_+]$.

We prove simultaneously that every v_k and every coefficient of f vanishes. Suppose this is known below order $m > 0$, and write $[S^m]f = c_m$. The coefficient of S^m in (3.8) is

$$(zm\,\mathrm{id} + D \cup -)v_m + mc_m[p_+] = 0. \tag{3.10}$$

If $v_m \neq 0$, the highest power of z in zmv_m cannot be cancelled by either $D \cup v_m$ or the constant term $mc_m[p_+]$, because v_m is polynomial in z. Hence $v_m = 0$, and then $c_m = 0$. Induction kills the complete tail and the possible unit part of the receiver coordinate, proving (3.3). □

Warning 3.2. Theorem 3.1 does not assert that the centre coordinates of $\Phi([p_-])$ vanish. The continuous Fourier map applies stationary phase to the transformed master fundamental solution. Positive master-Novikov terms can survive even though the classical restriction $a_p|_{F_0}$ is zero. The argument proves the ambient coordinate exactly because the wall-frequency derivative dies in the specialized completed source; polynomiality from Lemma 5.8 then controls the possible receiver unit correction.

The following consequence isolates the extra analytic premise required to turn (3.3) into a Gamma-row statement.

Hypothesis 3.3 (One-wall sectorial realization)**.** At a fixed nonzero wall parameter there is a pairing-preserving sectorial realization of (3.1) on one common admissible sector such that:

(a) the ambient formal summand realizes the intrinsic sectorial solution space of X_+;

(b) the wall summands realize the Gamma spans of the wall functors in the Shen–Shoemaker semiorthogonal decomposition;

(c) the realized point row has ambient coordinate (3.3).

Corollary 3.4 (Conditional wall-local rank identity)**.** *Under Hypothesis 3.3, the point row restricts to the point row of X_+ on the ambient packet and vanishes on every wall packet. In particular, for the whole generalized ζ_6 packet in this one receiver,*

$$b(X_-, \mathcal{P}_6(X_-)) = b(X_+, \mathcal{P}_6(X_+)). \tag{3.11}$$

Proof. The ambient assertion is Theorem 3.1 together with the pairing-preserving realization. A wall functor has image supported on the exceptional locus. Pairing its Gamma class with the point class in the common open gives zero Euler characteristic. Shen–Shoemaker's oriented Gamma asymptotics identify these supported classes with the wall blocks on the chosen common sector [SS25]. Hence the point row vanishes there. Restricting the resulting whole-block identity to the generalized ζ_6 packet gives (3.11). □

The hypothesis is not a reformulation of the cited formal theorems. Gu–Yu–Yu work over an exceptional Laurent completion; Shen–Shoemaker's asymptotic theorem is extremal and sectorial. Identifying the intrinsic large-radius point section with the full ambient packet across those frames is additional data.

4 A path-local theorem for ordinary flops

Let $Y \dashrightarrow Y'$ be a projective ordinary flop covered by [LLQW16], and let ℓ, ℓ' be its effective extremal fibre classes. The graph correspondence F identifies the big quantum products and Poincare pairings after analytic continuation

$$q' = q^{-1}, \qquad F(\ell) = -\ell'. \tag{4.1}$$

The continuation is analytic in the extremal variable and formal in the other Novikov variables.

Choose corresponding points $p \in Y$ and $p' \in Y'$ outside the exceptional loci. Let A be an ample divisor on the common contraction and write D, D' for its pullbacks. Then $D \cdot \ell = D' \cdot \ell' = 0$, while D is positive on every effective class away from the extremal ray. We complete the transverse Novikov ring by D-degree.

Theorem 4.1 (Point-section preservation)**.** *On each fixed nonsingular continuation domain of [LLQW16], the graph gauge satisfies*

$$F(s_Y(\mathcal{O}_p)) = s_{Y'}(\mathcal{O}_{p'}). \tag{4.2}$$

Consequently it preserves the Gamma point row.

Proof. At transverse degree zero, every nonconstant stable map has pure extremal class and lies in the exceptional locus. It cannot meet the chosen point. All positive-extremal descendant invariants with the point insertion therefore vanish. The point columns on both sides are their common classical column for every nonsingular value of q: $\widehat{\Gamma}_Y \operatorname{ch}(\mathcal{O}_p)$ is a scalar multiple of the top-degree class $[p]$, on which higher Gamma and Chern-class terms vanish. Analytic continuation of this identically classical extremal column remains classical, and the graph correspondence sends $[p]$ to $[p']$.

Let

$$\delta = F(s_Y(\mathcal{O}_p)) - s_{Y'}(\mathcal{O}_{p'}).$$

It is jointly flat and has zero transverse constant term. Suppose δ_β is a nonzero coefficient of minimal positive D-degree. Horizontality for the Novikov derivation defined by D gives

$$((D \cdot \beta) \operatorname{id} + z^{-1}(D \star_q -))\delta_\beta = 0. \tag{4.3}$$

At transverse degree zero, quantum corrections to $D\star_q$ have pure extremal degree. The divisor axiom multiplies them by $D \cdot \ell = 0$. Thus $D\star_q = D \cup -$ in (4.3). Cup product by D is nilpotent and $D \cdot \beta > 0$, so the displayed operator is invertible over $\mathbf{C}((z))$. This contradiction kills the first possible coefficient. Induction proves (4.2). Flatness of the pairing then preserves the row (2.3). □

Corollary 4.2 (Packetwise Boolean). *Let $\mathcal{P} \subset \mathcal{S}(Y)$ be a sectorially realized formal-monodromy packet whose chosen realization is transported by the Lee–Lin–Qu–Wang graph gauge to a sectorially realized packet $\mathcal{P}' \subset \mathcal{S}(Y')$. On the fixed continuation domain,*

$$b(Y, \mathcal{P}) = b(Y', \mathcal{P}'). \tag{4.4}$$

Proof. The graph gauge preserves the quantum connection, grading, first Chern class, and Poincare pairing. It therefore transports the complete $z = 0$ formal type. Theorem 4.1 identifies the covector on the transported sectorial realization, and (4.4) follows. □

The theorem is narrower than a Gamma/Fourier–Mukai comparison for every K-class. Pure extremal curves need not miss the support of a general class, and the support computation in the first paragraph of the proof would fail. Nor does the argument assert phase independence outside the chosen continuation domain.

5 A minimal ambient-target Stokes shear

Set $\alpha = 1/6$ and consider the rank-two meromorphic connection

$$\frac{d}{dz}\begin{pmatrix} y_0 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 & z^{-2} \\ 0 & z^{-2} + \alpha z^{-1} \end{pmatrix} \begin{pmatrix} y_0 \\ y_1 \end{pmatrix}. \tag{5.1}$$

Proposition 5.1 (Incomplete-Gamma countermodel). *Connection (5.1) has a formal line of monodromy ζ_6 whose sectorial lift undergoes a nonzero Stokes shear into a monodromy-one line. The shear can be equipped with a flat integral hyperbolic pairing and can be tensored with $\mathbf{Q}[N]/(N^3)$ without disappearing. Hence formal monodromy, pairing, integrality, and a commuting length-three nilpotent tag do not determine the zero or nonzero restriction of a flat covector to the ζ_6 line.*

Proof. Two flat columns are

$$f_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad f_\alpha(z) = \begin{pmatrix} \Gamma(1-\alpha, 1/z) \\ z^\alpha e^{-1/z} \end{pmatrix}. \tag{5.2}$$

Indeed,

$$\frac{d}{dz}\Gamma(1-\alpha, 1/z) = z^{\alpha-2} e^{-1/z}, \tag{5.3}$$

and direct differentiation verifies both rows of (5.1). The formal factors are the trivial line and $e^{-1/z} z^\alpha$, whose formal monodromy is $e^{2\pi i \alpha} = \zeta_6$.

Analytic continuation of the incomplete Gamma function adds a nonzero multiple of the complete Gamma value. After rescaling f_α, the Stokes jump has the form

$$f_\alpha^+ = f_\alpha^- + f_0. \tag{5.4}$$

Choose a flat covector r with $r(f_0) = 1$ and $r(f_\alpha^-) = 0$. Then $r(f_\alpha^+) = 1$. Thus the Boolean restriction of r to the sectorial lift of the same formal line changes.

Let $A(z)$ be the matrix in (5.1) and double the system by its dual:

$$B(z) = \mathrm{diag}(A(z), -A(z)^t), \qquad J = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}. \tag{5.5}$$

Then $B^t J + JB = 0$. If the jump on the first summand is $S = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, the doubled jump $\mathrm{diag}(S, S^{-t})$ preserves J and the obvious integral lattice after adjoining ζ_6. Finally tensor with $\mathbf{Q}[N]/(N^3)$ and let the jump act trivially on this factor. It commutes with N and still has (5.4). This proves the assertion. □

The example also occurs in a flat confluence family. Replacing z by z/t gives the z-equation

$$\frac{dY}{dz} = \begin{pmatrix} 0 & tz^{-2} \\ 0 & tz^{-2} + \alpha z^{-1} \end{pmatrix} Y, \tag{5.6}$$

with compatible t-equation inherited from the pullback connection. The two exponential factors collide at $t = 0$, while the nonzero Stokes coefficient persists on the ramified punctured t-line. Formal information at the confluent fibre cannot recover its target orientation.

Remark 5.2. Proposition 5.1 is an analytic no-go theorem, not a geometric counterexample. We do not assert that (5.1) is the quantum connection of a smooth projective wall crossing. It proves only that a geometric proof must use information absent from the listed formal shadows.

6 What localized Fourier–Laplace forgets

The incomplete-Gamma jump has an elementary differential-algebraic shadow. For $g(x) = \Gamma(1-\alpha, x)$,

$$(\partial_x + 1 + \alpha/x)\partial_x g = 0. \tag{6.1}$$

Differentiation forgets the constant solution, and that constant is the additive datum in the continuation formula.

Let $D_t = \mathbf{C}\langle t, \partial_t\rangle$. Under algebraic Fourier transform,

$$\mathbf{C}[t] = D_t/D_t\partial_t \quad \longmapsto \quad \delta_0 = D_\tau/D_\tau\tau. \tag{6.2}$$

Localizing at τ kills δ_0. Localized partial Fourier–Laplace therefore cannot recover the extension constant in (6.1). The four-term GKZ/Gauss–Manin comparison in [RSSW21] uses the same nonconservativity.

Proposition 6.1 (Punctual Fourier corner)**.** *Let*

$$\delta_{00} = D_{\tau_1,\tau_2}/D_{\tau_1,\tau_2}(\tau_1, \tau_2).$$

Localization at either τ_i kills δ_{00}, whereas

$$\mathrm{FL}^{-1}(\delta_{00}) = \mathcal{O}_{\mathbf{A}^2} \tag{6.3}$$

up to shift and sign conventions. Thus an object supported at the intersection of two deleted Fourier boundaries can have generic-rank-one inverse transform.

Proof. Use the Weyl-algebra Fourier automorphism $x_i \mapsto -\partial_{\tau_i}$ and $\partial_{x_i} \mapsto \tau_i$. It exchanges $D_x/D_x(\partial_{x_1}, \partial_{x_2})$ with δ_{00}. Since τ_i acts nilpotently on the latter, inverting either coordinate kills it. The inverse transform is the constant module, which has generic rank one. □

Tensoring Proposition 6.1 with the ζ_6 rank-one connection, its hyperbolic dual, and $\mathbf{Q}[N]/(N^3)$ preserves every conclusion. Bare double-boundary support is therefore not a rank-zero condition.

Nor does marking one punctual vector make it unique. Let C be any finite-dimensional coefficient space and put $\mathcal{B} = \delta_{00} \otimes C$. For $G \in \mathrm{GL}(C)$, two receivers can differ by G while having the same support and characteristic cycle. After hyperbolic doubling, $G \oplus G^{-t}$ preserves a perfect pairing. It can still change a chosen point row. Consequently, a claim that the punctual quotient is generated by one marked line is additional geometry.

Enlarging the discarded subspace to the kernel of the point row makes the quotient one-dimensional only tautologically: invariance of that kernel is the theorem one seeks.

The unlocalized boundary map

$$\mathrm{FL}(M) \longrightarrow j_* j^* \,\mathrm{FL}(M), \qquad j : \mathbf{G}_m \hookrightarrow \mathbf{A}^1, \tag{6.4}$$

retains the missing object. In a two-coordinate problem one must also retain the orientation, duality, and $! \to *$ or can / var maps of the complete Cech boundary square. Proposition 6.1 shows that retaining the object is necessary, not that its support alone proves the desired vanishing.

7 The two-wall criterion

Let Y be an intermediate smooth projective variety, $D \subset Y$ the union of the two incident wall boundaries, and $p \in U := Y \setminus D$. Suppose a common analytic receiver contains the two sectorial realizations under comparison, and let T be their transition. Write $\mathcal{S}_D(Y)$ for a span of Gamma sections represented by classes supported on D. The point row annihilates this span:

$$\mathfrak{r}_{Y,p}(v) = 0 \qquad (v \in \mathcal{S}_D(Y)). \tag{7.1}$$

Hypothesis 7.1 (Rank-zero-target condition)**.** On the packet under consideration, the relative transition admits a canonical ray-ordered factorization

$$T = \prod_{a=1}^{m} (1 + v_a \otimes \lambda_a) \tag{7.2}$$

whose complete residue targets v_a belong to $\mathcal{S}_D(Y)$.

The word *complete* is essential. Refining a residue block into coordinate terms can create nonzero ranks which cancel only after the full block is assembled.

Theorem 7.2 (One-row assembly)**.** *Under Hypothesis 7.1,*

$$\mathfrak{r}_{Y,p} T = \mathfrak{r}_{Y,p}. \tag{7.3}$$

Consequently, any finite factorization whose ordinary-flop steps satisfy Theorem 4.1 and whose discrepant adjacent-wall transitions satisfy Hypothesis 7.1 preserves the point-row Boolean on the transported packet.

Proof. Equation (7.1) gives

$$\mathfrak{r}_{Y,p}(1 + v_a \otimes \lambda_a) = \mathfrak{r}_{Y,p}$$

for every factor in (7.2). Multiplying the identities proves (7.3). The factorization statement follows by induction, using Theorem 4.1 at crepant steps. □

A stronger geometric condition is one-sided kernel support. If the cone of the transition kernel relative to the diagonal has support

$$\mathrm{Supp}(K_{\mathrm{rel}}) \subset Y \times D, \tag{7.4}$$

with D in the output factor, and if its induced action satisfies $\mathrm{im}(T - \mathrm{id}) \subset \mathcal{S}_D(Y)$, then (7.1) gives $\mathfrak{r}_{Y,p} T = \mathfrak{r}_{Y,p}$ directly. This conclusion does not require, and does not imply, that every factor in an independently chosen factorization of T has supported target. Support in $D \times Y$ instead constrains the source and would not imply (7.3).

There is a tempting singular shadow of the factorwise condition. Let B_{corner} be a complete oriented correction left by the two localizations after subtracting the common point contribution, and let its signed punctual multiplicity be the alternating δ_{00} coefficient

$$\mu_{00}(B_{\mathrm{corner}}).$$

Fourier transform exchanges a punctual coefficient with a zero-section multiplicity, hence with generic output rank in the elementary model of Proposition 6.1. One might therefore hope for

$$\mu_{00}(B_{\mathrm{corner}}) = 0 \tag{7.5}$$

as a numerical test. This implication is *not* proved here. A single alternating multiplicity can vanish by cancellation even when individual ray targets have nonzero point row, and no exact point-row marking on B_{corner} has yet been constructed. Proposition 6.1 also explains why merely knowing that the boundary object is punctual proves nothing.

The failure of a global-to-factorwise inference already occurs in dimension two. Let $r = e_1^*$ on $V = \mathbf{C}^2$, and put $A = e_1 \otimes e_2^*$. Then $A^2 = 0$ and

$$\mathrm{id} = (\mathrm{id} + A)(\mathrm{id} - A), \tag{7.6}$$

although both elementary factors have target e_1 and $r(e_1) = 1$. Thus even a vanishing total correction does not force the targets in a chosen ray factorization to lie in $\ker r$.

The missing comparison can be stated blockwise: for each complete ray block one would need an unlocalized boundary object B_a and an identity

$$\mathfrak{r}_{Y,p}(v_a) = \mu_{00}(B_a). \tag{7.7}$$

with enough concentration or effectivity to prevent cancellation inside the block. Equation (7.7) and $\mu_{00}(B_a) = 0$ for every a would imply Hypothesis 7.1; the single global equation (7.5) does not.

Remark 7.3 (What remains to be proved). Neither (7.2), the action condition following (7.4), nor the blockwise comparison (7.7) is deduced here for a general pair of discrepant walls. Existing one-wall theorems determine formal decompositions and, in special toric settings, Gamma/window comparisons. The missing statement is directed and two-wall: it must retain which side of every vanishing cycle is the output and must compare that orientation with the Gamma point row.

8 Scope, source boundaries, and the next problem

The results have deliberately different logical strengths.

(1) Theorem 3.1 is an exact identity inside the Gu–Yu–Yu completed formal comparison. It concerns the ambient point coordinate only.

(2) Theorem 4.1 is an intrinsic point-section identity on one fixed Lee–Lin–Qu–Wang continuation domain. It does not assert full descendant invariance or a Gamma/Fourier–Mukai square for all classes.

(3) Corollary 3.4 requires the named one-wall sectorial realization hypothesis. The formal Gu–Yu–Yu decomposition and the extremal Shen–Shoemaker asymptotics do not by themselves identify the intrinsic large-radius point section in a common receiver.

(4) Propositions 5.1 and 6.1 are exact analytic and algebraic countermodels. They are not asserted to arise from smooth projective quantum connections.

(5) Theorem 7.2 is a conditional assembly theorem. Its proof is exact once the directed target-support hypothesis is supplied.

There is a second, geometric limitation. Birational cobordism decomposes a birational map into elementary toroidal wall crossings, but smooth master spaces can have chamber quotients with finite-quotient singularities [Wlo00, AKMW02]. Gu–Yu–Yu assume smooth projective chamber quotients. Resolving the singular chambers introduces further walls and does not provide the missing common sectorial point-row realization for free. The paper therefore makes no general factorization theorem without Hypothesis 7.1.

The boundary-loss mechanism also limits what a GKZ argument can establish. For quasi-symmetric representations, the GKZ local system can carry a categorical window marking [SVdB22]. Discrepant confluence to the irregular problem uses localized Fourier–Laplace transforms, which kill the constant modules in (6.2). An unlocalized comparison with its boundary triangle is needed before that marking can control the point row. General discrepant toric continuation results such as [AS20] do not, without additional Gamma calibration, supply the two-wall identity (7.5).

The resulting open problem is precise.

> For two incident noncrepant unit wall crossings with a common smooth projective chamber, construct the complete oriented Fourier/window boundary correction block by block, construct the point-row marking (7.7), and prove that every marked block has zero punctual coefficient, or exhibit a geometric residue block for which it does not.

This asks for one numerical row rather than a full equality of Stokes and window matrices. The countermodels show that the row cannot be recovered from formal monodromy, a flat pairing, integrality, a nilpotent divisor tag, or unmarked singular support. A proof must retain the augmentation marking through the unlocalized two-wall comparison.

AI assistance disclosure

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